TM Forum Specification

Product Usage Catalog Management User Guide

TMF767

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Introduction

This document is the user guide of the REST API for Product Usage Catalog Management. It includes the data model definition as well as all available operations for the ProductUsageSpecification resource.

The Product Usage Catalog Management API provides a standardized mechanism for Product Usage Catalog management such as the creation, update, retrieval of a collection of Product Usage Specification. A Product Usage Specification defines how usage details are collected for a Product that is charged or metered for usage (for example a mobile voice product). A Product Usage Specification is a catalog representation of a usage of a Product; it defines the characteristics of the usage associated with a Product Specification.

The Product Usage Catalog API manages the ProductUsageSpecification resource:

- A ProductUsageSpecification represents a catalog view of the targeted ProductUsage, comprised of characteristics specifications which define all the attributes known for that particular type of usage.
- The main attributes are its id, lastUpdate, related ProductUsageSpecification, characteristics specifications, lifecycleStatus
- A ProductUsageSpecification follows a simple lifecycle which is presented below.

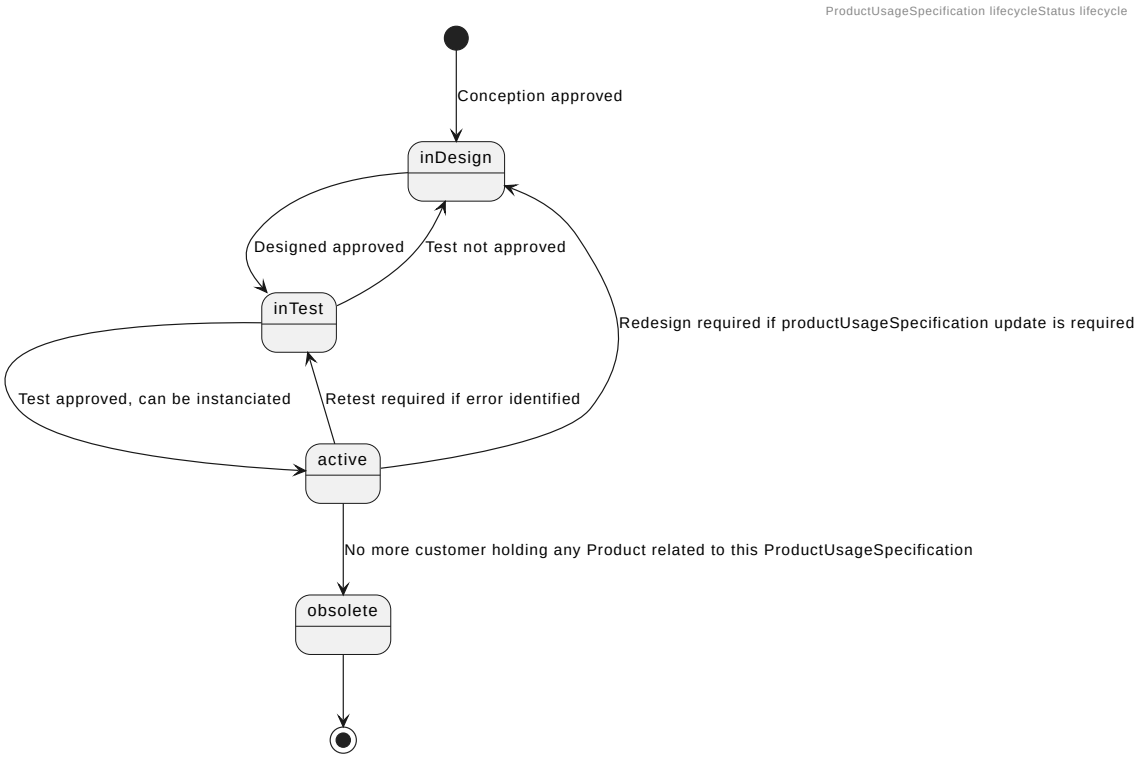
The Product Usage Catalog API performs the following operations on the ProductUsageSpecification resource:

- Creation of a ProductUsageSpecification
- Retrieval of a ProductUsageSpecification or a collection of ProductUsageSpecification, depending on filter criteria
- Partial update of a ProductUsageSpecification
- Deletion of a ProductUsageSpecification

Note : TMF635 Product Usage Management API manages ProductUsage events that represents the structured report of a service that has been delivered, guided to a product, and is considered billable. Each ProductUsage event is defined by a corresponding ProductUsageSpecification, which standardizes what type of usage occurred (e.g., voice, data session, API call) and its associated characteristics (e.g., duration, data consumption, correlation identifier). TMF767 Product Usage Catalog Management API is responsible for managing ProductUsageSpecifications that represents a catalog view of the targeted ProductUsage comprised of characteristics specifications which define all the attributes know for the particular usage

ProductUsageSpecification lifecycleStatus

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State	Definition
inDesign	When the conception of the ProductUsageSpecification is accepted
inTest	When the design is approved
active	When the test is OK then its status is changed to “active”. When a ProductUsageSpecification is in a “active” status it means, it has been validated and tested, and can be referred by a ProductSpecification available for customers.
obsolete	When no more customer holds any Product being an instantiation of a ProductSpecification referring to this ProductUsageSpecification, its status is changed to “obsolete” meaning it can be removed from the catalog.

Information Framework

This TMF767 Product Usage Catalog API manages entities that are described in Information Framework (SID) in Product Usage ABE as ProductUsageSpec entity.

These links permit to retrieve Information Framework documentation regarding Product Usage domain :

- Usage Specification : https://www.tmforum.org/MODA/EARoot/EA4/EA2/EA4/EA6/EA1712.htm
- Product Usage Part 1 : https://www.tmforum.org/MODA/EARoot/EA4/EA2/EA4/EA6/EA1716.htm

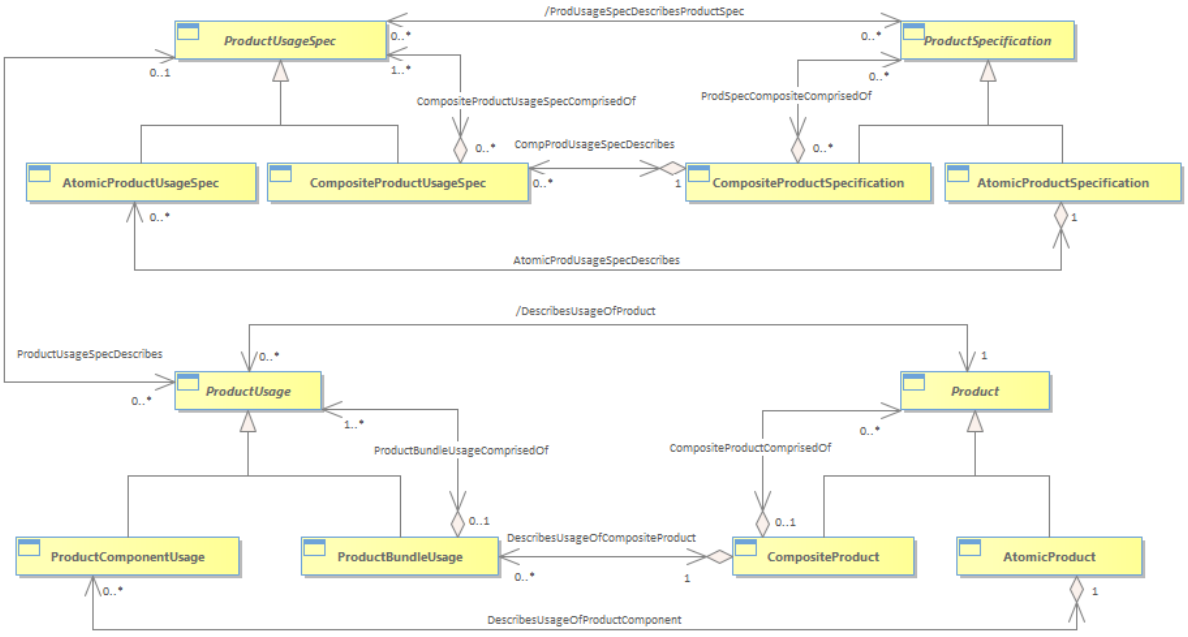
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- Product Usage Part 2 : https://www.tmforum.org/MODA/EARoot/EA4/EA2/EA4/EA6/EA1718.htm

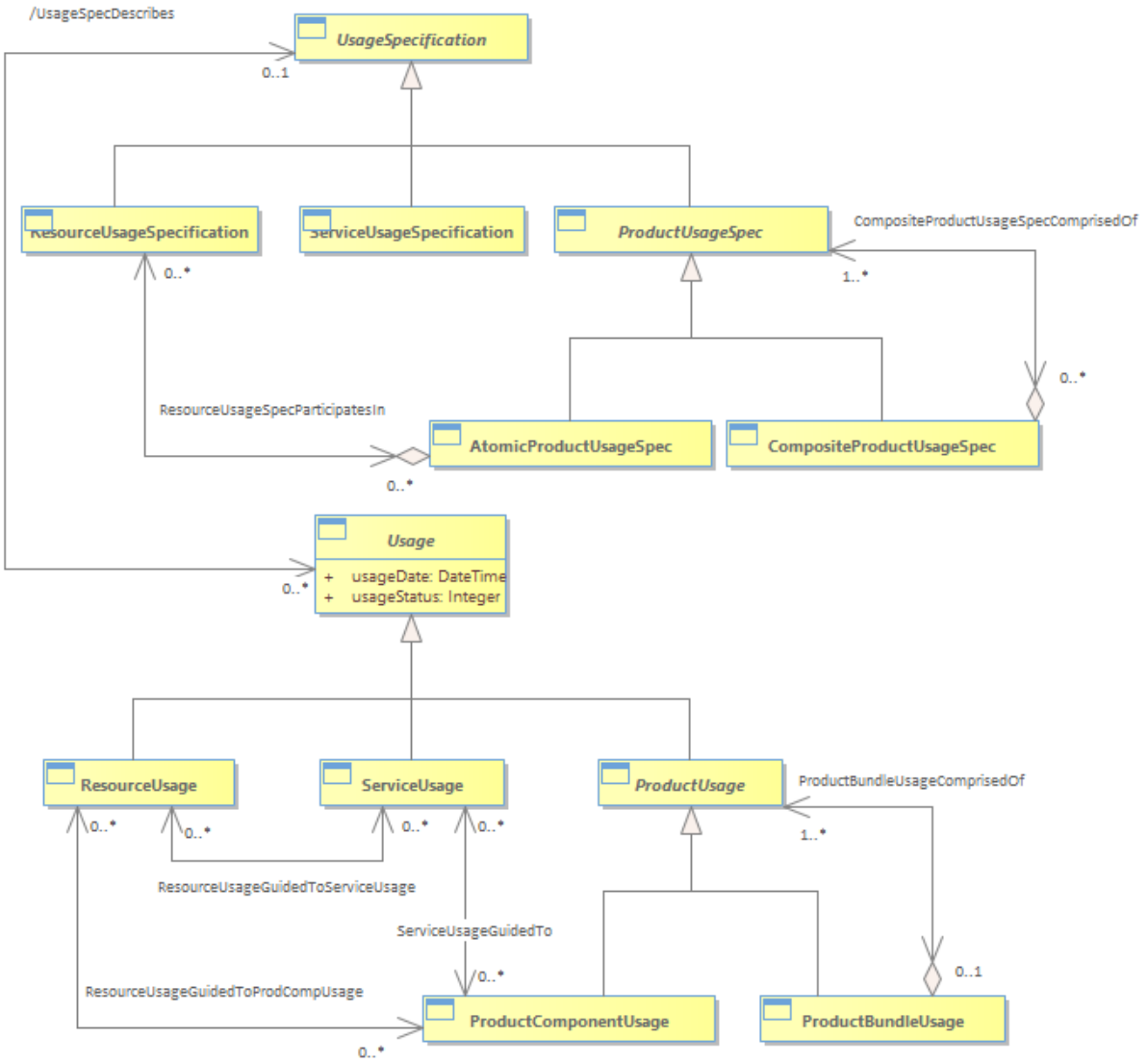
Information Framework excerpts:

Put simply, usage is how much service is used, by whom is it used, where and when is it used and circumstances under which it is used.

The UsageSpecification class describes a type of usage. In order to achieve a flexible structure of the UsageSpecification all its attributes are stored as characteristics.



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The Usage class represents a usage event. It is described by the UsageSpecification and comprised of UsageCharacteristicValues that hold data attributes of a usage specified by UsageSpecCharacteristics. The UsageCharacteristic can either store its value in the usageCharValue property or reference predefined UsageSpecCharacteristicValue.

The ProductUsageSpec is defined for the certain ProductSpecification.

Note that the API representation of Information Framework model is partial. In particular the direct relationship between a ProductUsage (that would be originated from a ResourceUsage) and a ResourceUsage is not implemented because of lack of business need for this.

Note that Information Framework and API specification may evolve independently but tend to be consistent as much as possible.

ODA Components

This TMF767 Product Usage Catalog API is exposed by TMFC001 Product Catalog Management. It exposes ProductUsageSpecification resource, at least with GET operation towards other components.

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Note that ODA Component and API specification may evolve independently but tend to be consistent as much as possible.

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Sample Use Cases

Use Cases

Main use cases for the Product Usage Catalog API are :

- The creation of a product usage specification as part of product development of a new service
- The management of existing product usage specifications as part of changes to services refected in the product level
- The association of product usage specifications in the definition of product specifications
- Retrieval of the product usage specifications to be able to understand the characteristics of the product usages

The TMFS012 Product Catalog – Launching a New Product Offering should cover the use of this API. In its existing version it doesn't yet, that's why there are no excerpts of this use case in this document yet. The reader can still have a look on it as the document will evolve to cover this API. (https://www.tmforum.org/resources/technical-specification/tmfs012-use-case-product-catalog-launching-a-new-product-offering-v2-0-2/)

Currently the examples in this User Guide and OAS file come from the TMFS005 Use Case Billing Fiber Contract (https://www.tmforum.org/resources/technical-specification/tmfs005-billing-fiber-contract-v2-2-0/) and TMFS009 Use Case Usage and Balance Management (https://www.tmforum.org/resources/technical-specification/tmfs009-use-case-usage-and-balance-management-v6-1-0/) documents.

Note that these two url will propose latest version of the documents.

Note that TMFS005 and TMFS009 show the data model but it doesn't really provide the use case for this API. Those two documents are both more related to the creation of the actual ProductUsage in the inventory and where the ProductUsageSpecification plays the role of explaining the characteristics used for the given service.

- Creation of a simple ProductUsageSpecification for International Voice of a Fix Communication ProductSpecification. This ProductUsageSpecification is defined by some characteristics of a Fix Communication voice call (originating phone number, destination phone number, start and stop date-and-time…); listing ProductUsageSpecification's at version 2.5; retrieving a single ProductUsageSpecification based on its id; updating the ProductUsageSpecification for a simple change; and ultimately its deletion based on the same id.

Note that entities and characteristic names don't always fully match but the meaning should match.

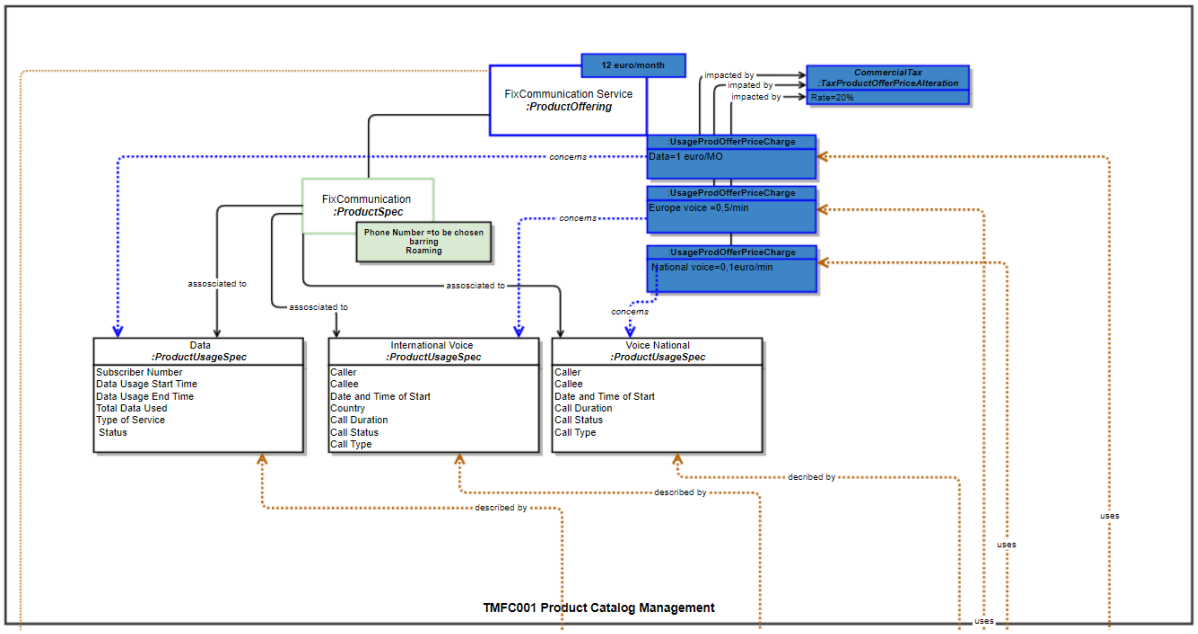
Catalog view

This excerpt from TMFS005 presents the catalog view of a Fix Communication offering.

The Fix Communication ProductSpecification has 3 ProductUsageSpecification : National Voice, International Voice and Data. These ProductUsageSpecification are under the

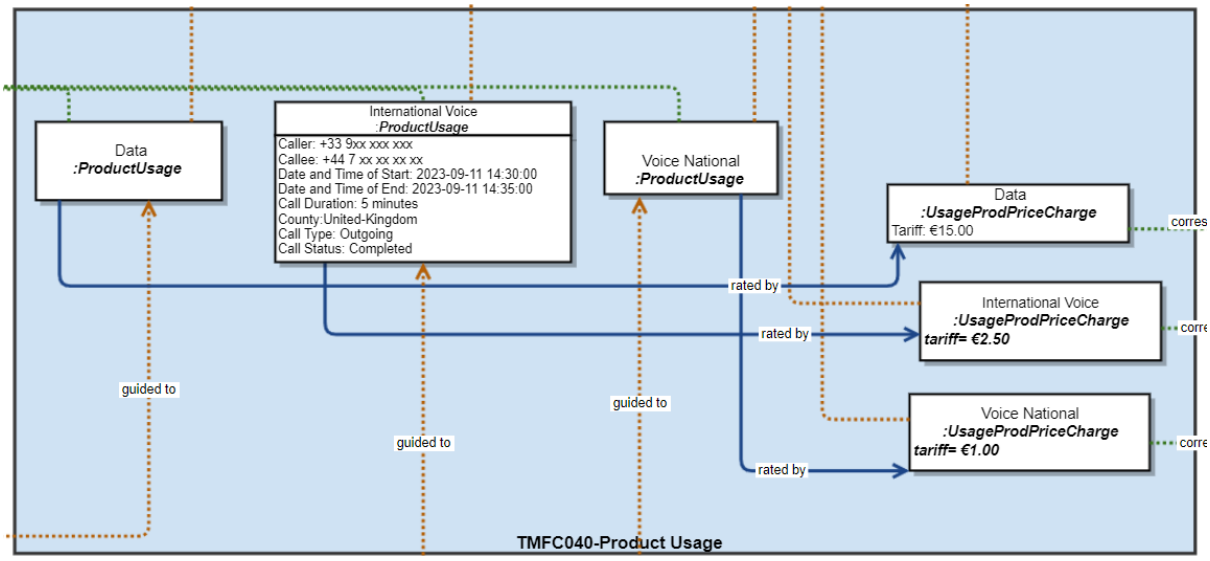
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responsibility of the Product Catalog component. They can be created and updated by a employee of a marketing team having access to the Product Catalog administration software.



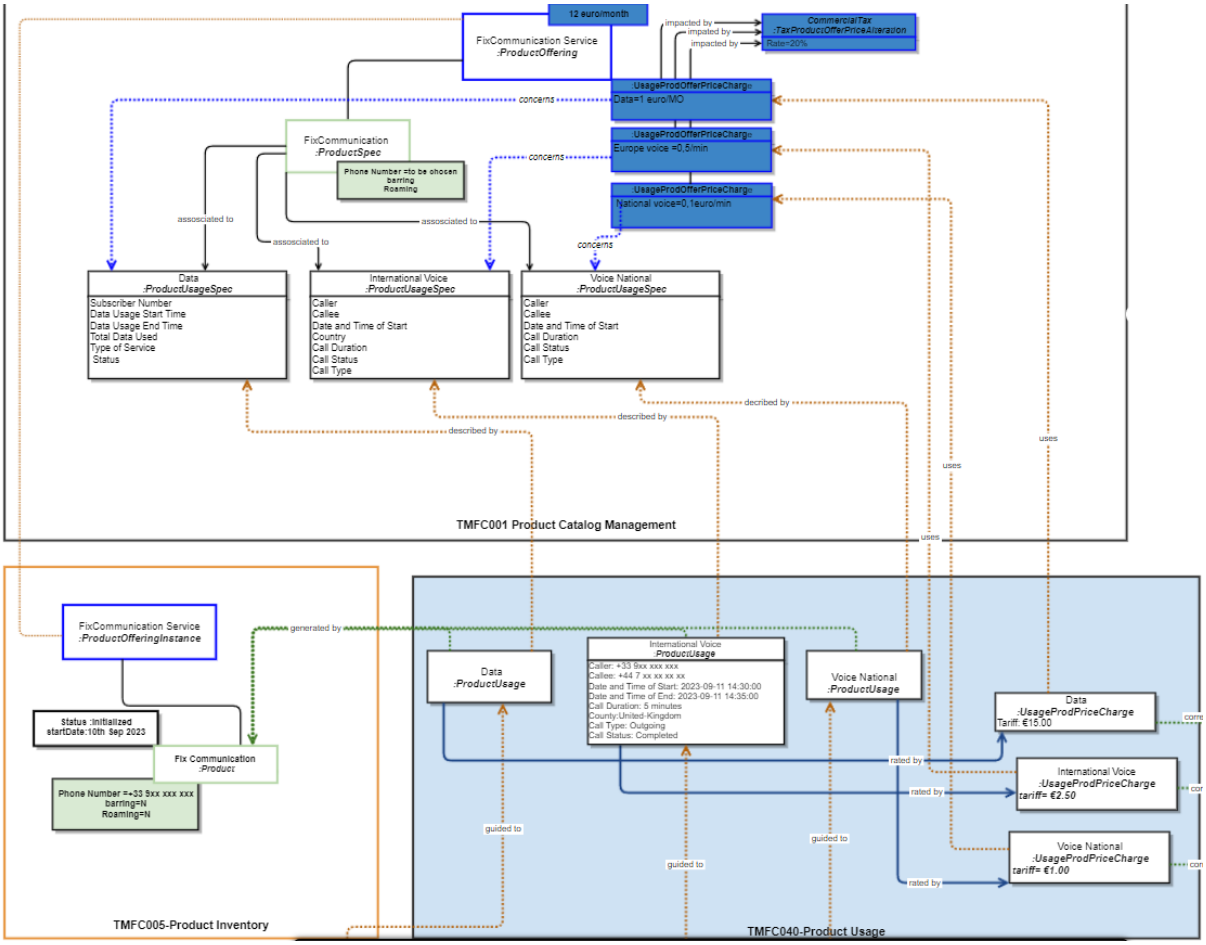
Inventory view

This excerpt from TMFS005 presents the ProductUsage representation once created and rated.

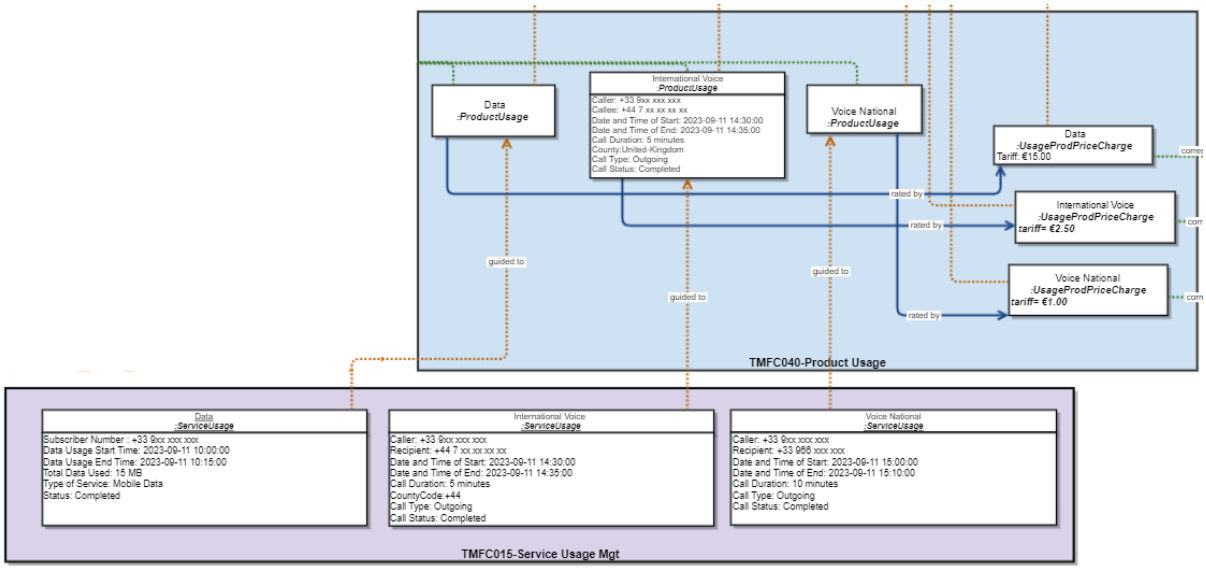


This view permits to see the relationship between the inventory view of the Fix Communication product and its associated ProductUsage created after a voice call has been completed, and also the relationship between the ProductUsage and their corresponding ProductUsageSpecification, same for UsageProductPrice and ProductOfferingPrice.

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These ProductUsage are created based on preliminary ServiceUsage entities that contain characteristics of the voice call.



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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a StringCharacteristic and BooleanCharacteristic inheriting properties from the abstract Characteristic entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Characteristic instances some may be instances of StringCharacteristic where other could be instances of BooleanCharacteristic. The @type gives this information. All resources and sub-resources of the Product Usage API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance a ProductUsageSpecificationRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as ProductUsageSpecificationRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

ProductUsageSpecification resource

A detailed description of a product usage specification that are of interest to the business and can have product offering prices applied to it. It is comprised of characteristic specification, which define all attributes and their possible values known for a particular type of product usage.

Resource model

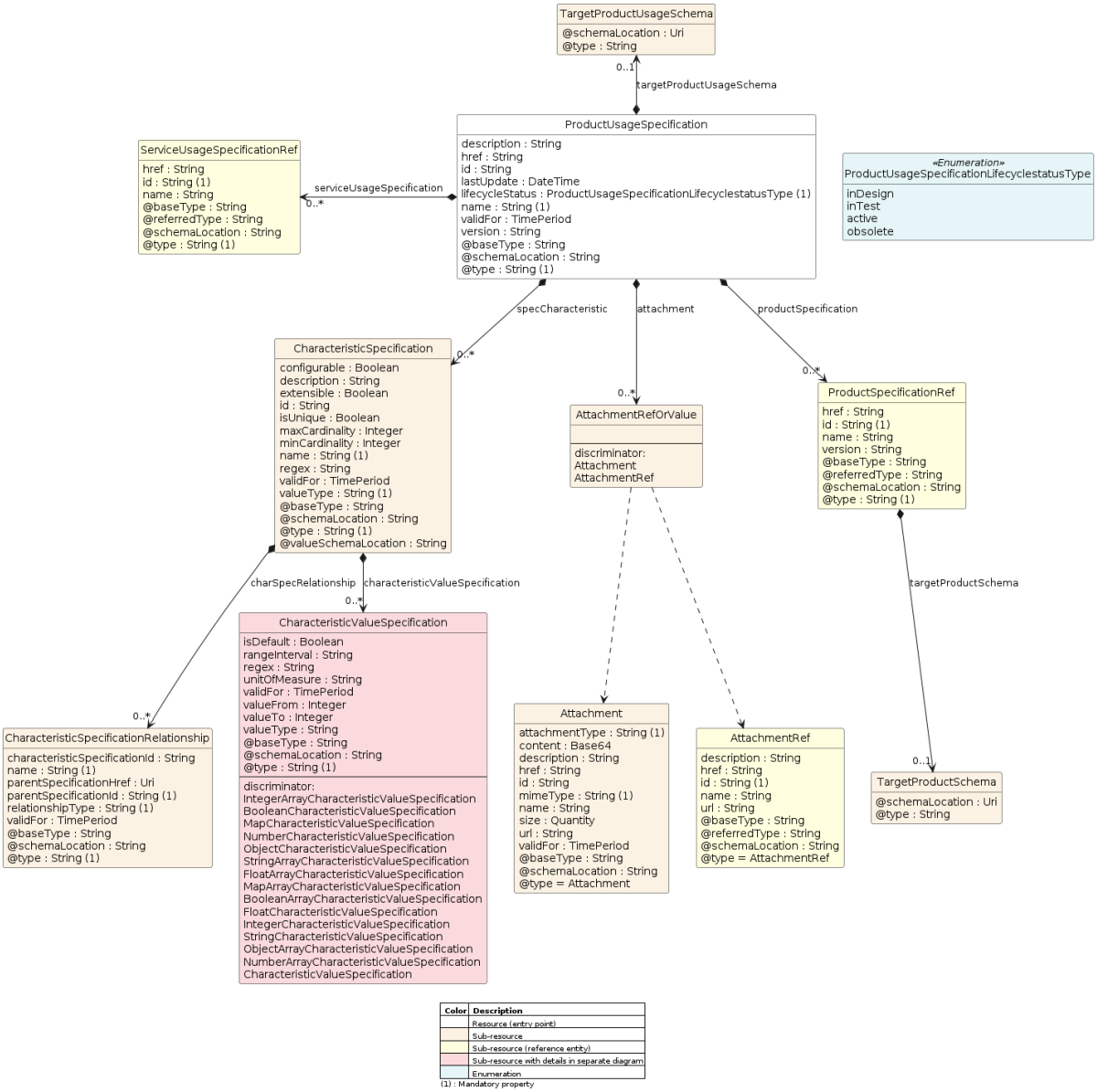


Figure 1 - ProductUsageSpecification

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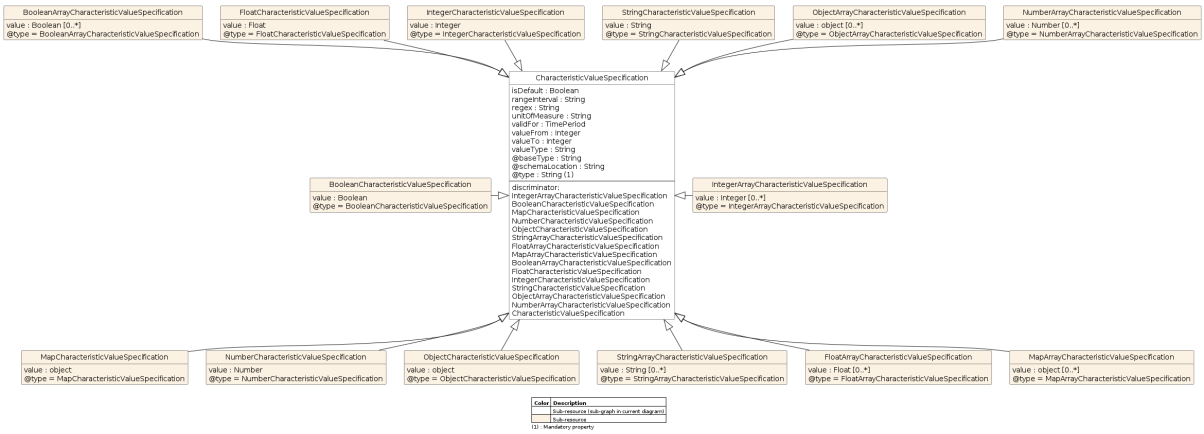


Figure 2 - CharacteristicValueSpecification

Field descriptions

ProductUsageSpecification fields

attachment	An AttachmentRefOrValue. Attachments that may be of relevance to this specification, such as picture, document, media.
description	A String. Description of the product usage specification.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. Date and time of the last update of the specification.
lifecycleStatus	A ProductUsageSpecificationLifecyclestatusType. Possible values for the lifecycle status of the product usage specification. Refer to User Guide for comprehensive lifecycle schema and descriptions of status. ENUMERATED with values: * inDesign * inTest * active * obsolete
name	A String. Name given to the product usage specification.
productSpecification	A ProductSpecificationRef. Relationship to the product specifications that use this product usage specification to rate or meter their products.
serviceUsageSpecification	A ServiceUsageSpecificationRef. List of service usage specification(s) that are involved for the creation of this product usage specification.
specCharacteristic	A CharacteristicSpecification. List of characteristic specifications that will be instantiated at the product usages creation.
targetProductUsageSchema	A TargetProductUsageSchema. The reference object to the schema and type of target product usage which is described by product usage specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

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version	A String. Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

attachmentType	A String. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
targetProductSchema	A TargetProductSchema. The reference object to the schema and type of target product which is described by product specification.
version	A String. Version of the product specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TargetProductSchema sub-resource fields

@schemaLocation	A Uri. This field provides a link to the schema describing the target product.
@type	A String. Class type of the target product.

ServiceUsageSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicSpecification sub-resource fields

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charSpecRelationship	A CharacteristicSpecificationRelationship. An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
characteristicValueSpecification	A CharacteristicValueSpecification. A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a CharacteristicSpecification object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding Characteristic object can take on. CharacteristicValueSpecification can be instantiated as * BooleanArrayCharacteristicValueSpecification * BooleanCharacteristicValueSpecification * FloatArrayCharacteristicValueSpecification * FloatCharacteristicValueSpecification * IntegerArrayCharacteristicValueSpecification * IntegerCharacteristicValueSpecification * MapArrayCharacteristicValueSpecification * MapCharacteristicValueSpecification * NumberArrayCharacteristicValueSpecification * NumberCharacteristicValueSpecification * ObjectArrayCharacteristicValueSpecification * ObjectCharacteristicValueSpecification * StringArrayCharacteristicValueSpecification * StringCharacteristicValueSpecification
configurable	A Boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A String. A narrative that explains the CharacteristicSpecification.
extensible	A Boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A String. Unique ID for the characteristic.
isUnique	A Boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An Integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An Integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A String. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.

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regex	A String. A rule or principle represented in regular expression used to derive the value of a characteristic value.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueType	A String. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
@valueSchemaLocation	A String. This (optional) field provides a link to the schema describing the value type.

CharacteristicSpecificationRelationship sub-resource fields

characteristicSpecificationId	A String. Unique identifier of the characteristic within the specification.
name	A String. Name of the target characteristic within the specification.
parentSpecificationHref	A Uri. Hyperlink reference to the parent specification containing the target characteristic.
parentSpecificationId	A String. Unique identifier of the parent specification containing the target characteristic.
relationshipType	A String. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

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unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	CharacteristicValueSpecification can be instantiated as * BooleanArrayCharacteristicValueSpecification * BooleanCharacteristicValueSpecification * FloatArrayCharacteristicValueSpecification * FloatCharacteristicValueSpecification * IntegerArrayCharacteristicValueSpecification * IntegerCharacteristicValueSpecification * MapArrayCharacteristicValueSpecification * MapCharacteristicValueSpecification * NumberArrayCharacteristicValueSpecification * NumberCharacteristicValueSpecification * ObjectArrayCharacteristicValueSpecification * ObjectCharacteristicValueSpecification * StringArrayCharacteristicValueSpecification * StringCharacteristicValueSpecification
value	This property is present in subclasses

BooleanArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.

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validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Boolean.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Boolean. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristicValueSpecification sub-resource fields

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isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Float.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Float. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.

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valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An Integer.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

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unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An Integer. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MapArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An object.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

MapCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An object. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

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value	A Number.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Number. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.

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rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An object. Collection of characteristic values.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An object. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

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unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TargetProductUsageSchema sub-resource fields

@schemaLocation	A Uri. This field provides a link to the schema describing the target product usage.
@type	A String. Class type of the target product usage.

Json representation sample(s)

We provide below a JSON representation as example of the 'ProductUsageSpecification' resource object.

```
{
  "@type": "ProductUsageSpecification",
  "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
  "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
  "name": "InternationalVoice",
  "description": "Specification for international voice calls usage",
  "version": "2.5.0",
  "lastUpdate": "2022-05-20T19:00:00Z",
  "lifecycleStatus": "inDesign",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the caller",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
```

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```
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the caller, must start with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the called party",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the called party, must start
with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "duration",
      "valueType": "integer",
      "description": "Total duration of the call, in seconds.",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "startDateTime",
      "valueType": "date-time",
      "description": "startdate and starttime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "endDateTime",
      "valueType": "date-time",
      "description": "enddate and endtime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callType",
      "valueType": "string",
      "description": "Type of the call : outgoing or incoming",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
```


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```
            "value": "outgoing"
          },
          {
            "@type": "StringCharacteristicValueSpecification",
            "valueType": "string",
            "value": "incoming"
          }
        ]
      },
      {
        "@type": "CharacteristicSpecification",
        "name": "callStatus",
        "valueType": "string",
        "description": "Status of the call : failed or completed",
        "minCardinality": 1,
        "maxCardinality": 1,
        "characteristicValueSpecification": [
          {
            "@type": "StringCharacteristicValueSpecification",
            "valueType": "string",
            "value": "failed"
          },
          {
            "@type": "StringCharacteristicValueSpecification",
            "valueType": "string",
            "value": "completed"
          }
        ]
      }
    ],
    "validFor": {
      "startDateTime": "2022-05-20T19:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
    },
    "productSpecification": [
      {
        "@type": "ProductSpecificationRef",
        "@referredType": "ProductSpecification",
        "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productSpecification/cbfa3598-2276-4b35-8f7f-
3252d9b8f397",
        "id": "cbfa3598-2276-4b35-8f7f-3252d9b8f397",
        "name": "FixCommunication",
        "version": "1.0"
      }
    ],
    "serviceUsageSpecification": [
      {
        "@type": "ServiceUsageSpecificationRef",
        "@referredType": "ServiceUsageSpecification",
        "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsageSpecification/6d414e5c-6e49-4651-b4a9-
64f698bda943",
        "id": "6d414e5c-6e49-4651-b4a9-64f698bda943",
        "name": "InternationalVoice"
      }
    ]
}
```


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Notification Resource Models

4 notifications are defined for this API.

Notifications related to ProductUsageSpecification:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

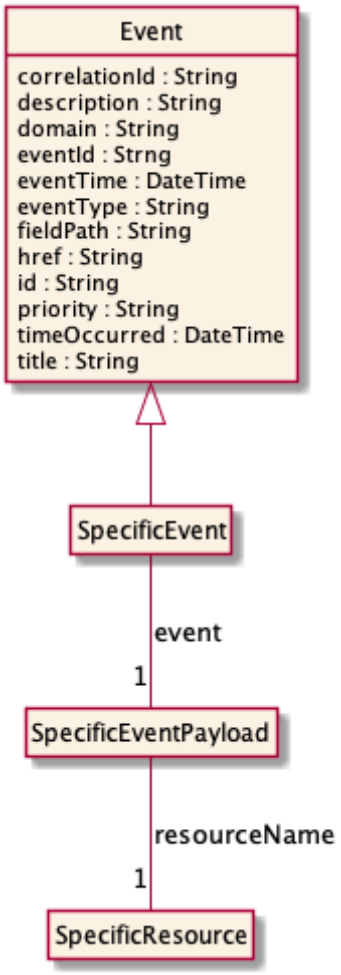


Figure 3 Notification Pattern

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ProductUsageSpecification

Create Event

Creating a ProductUsageSpecificationCreateEvent event

```
Content-Type: application/json
{
  "@type": "ProductUsageSpecificationCreateEvent",
  "@baseType": "Event",
  "domain": "Product Usage",
  "timeOccurred": "2022-05-20T19:00:00Z",
  "eventId": "60860fce-f1ea-4038-9503-57451b8b5a5d",
  "eventType": "ProductUsageSpecificationCreateEvent",
  "event": {
    "productUsageSpecification": {
      "@type": "ProductUsageSpecification",
      "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
      "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
      "name": "InternationalVoice",
      "description": "Specification for international voice calls usage",
      "version": "2.5.0",
      "lastUpdate": "2022-05-20T19:00:00Z",
      "lifecycleStatus": "inDesign",
      "specCharacteristic": [
        {
          "@type": "CharacteristicSpecification",
          "name": "originatingCountryCode",
          "valueType": "string",
          "description": "ISO 3166 country code of the caller",
          "regex": "^[A-Z]{2}$",
          "minCardinality": 1,
          "maxCardinality": 1
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "originatingPhoneNumber",
          "valueType": "string",
          "description": "International phone number of the caller, must start with
+",
          "regex": "^\\+[1-9]\\d{1,14}$",
          "minCardinality": 1,
          "maxCardinality": 1
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "destinationCountryCode",
          "valueType": "string",
          "description": "ISO 3166 country code of the called party",
          "regex": "^[A-Z]{2}$",
          "minCardinality": 1,
          "maxCardinality": 1
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "destinationPhoneNumber",
          "valueType": "string",
          "description": "International phone number of the called party, must start
with +",
```

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```
            "regex": "^\\+[1-9]\\d{1,14}$",
            "minCardinality": 1,
            "maxCardinality": 1
          },
          {
            "@type": "CharacteristicSpecification",
            "name": "duration",
            "valueType": "integer",
            "description": "Total duration of the call, in seconds.",
            "minCardinality": 1,
            "maxCardinality": 1
          },
          {
            "@type": "CharacteristicSpecification",
            "name": "startDateTime",
            "valueType": "date-time",
            "description": "startdate and starttime of the call, in date-time format",
            "minCardinality": 1,
            "maxCardinality": 1
          },
          {
            "@type": "CharacteristicSpecification",
            "name": "endDateTime",
            "valueType": "date-time",
            "description": "enddate and endtime of the call, in date-time format",
            "minCardinality": 1,
            "maxCardinality": 1
          },
          {
            "@type": "CharacteristicSpecification",
            "name": "callType",
            "valueType": "string",
            "description": "Type of the call : outgoing or incoming",
            "minCardinality": 1,
            "maxCardinality": 1,
            "characteristicValueSpecification": [
              {
                "@type": "StringCharacteristicValueSpecification",
                "valueType": "string",
                "value": "outgoing"
              },
              {
                "@type": "StringCharacteristicValueSpecification",
                "valueType": "string",
                "value": "incoming"
              }
            ]
          },
          {
            "@type": "CharacteristicSpecification",
            "name": "callStatus",
            "valueType": "string",
            "description": "Status of the call : failed or completed",
            "minCardinality": 1,
            "maxCardinality": 1,
            "characteristicValueSpecification": [
              {
                "@type": "StringCharacteristicValueSpecification",
                "valueType": "string",
                "value": "failed"
              },
              {
                "@type": "StringCharacteristicValueSpecification",
```


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```
              "valueType": "string",
              "value": "completed"
            }
          ]
        }
      ],
      "validFor": {
        "startDateTime": "2022-05-20T19:00:00Z",
        "endDateTime": "2099-01-11T00:00:00Z"
      },
      "productSpecification": [
        {
          "@type": "ProductSpecificationRef",
          "@referredType": "ProductSpecification",
          "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productSpecification/cbfa3598-2276-4b35-8f7f-
3252d9b8f397",
          "id": "cbfa3598-2276-4b35-8f7f-3252d9b8f397",
          "name": "FixCommunication",
          "version": "1.0"
        }
      ],
      "serviceUsageSpecification": [
        {
          "@type": "ServiceUsageSpecificationRef",
          "@referredType": "ServiceUsageSpecification",
          "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsageSpecification/6d414e5c-6e49-4651-b4a9-
64f698bda943",
          "id": "6d414e5c-6e49-4651-b4a9-64f698bda943",
          "name": "InternationalVoice"
        }
      ]
    }
  }
}
```


Delete Event

Creating a ProductUsageSpecificationDeleteEvent event


```
Content-Type: application/json
{
  "@type": "ProductUsageSpecificationDeleteEvent",
  "@baseType": "Event",
  "domain": "Product Usage",
  "timeOccurred": "2022-12-31T23:00:00Z",
  "eventId": "71adedc7-5b64-419a-8772-d9af59394665",
  "eventType": "ProductUsageSpecificationDeleteEvent",
  "event": {
    "productUsageSpecification": {
      "@type": "ProductUsageSpecification",
      "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
      "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f"
    }
  }
}
```


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Attribute Value Change Event

Creating a ProductUsageSpecificationAttributeValueChangeEvent event

```
Content-Type: application/json
{
  "@type": "ProductUsageSpecificationAttributeValueChangeEvent",
  "@baseType": "Event",
  "domain": "Product Usage",
  "timeOccurred": "2022-05-30T10:00:00Z",
  "eventId": "1fd7195e-9831-4823-947f-f49665daced7",
  "eventType": "ProductUsageSpecificationAttributeValueChangeEvent",
  "event": {
    "productUsageSpecification": {
      "@type": "ProductUsageSpecification",
      "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
      "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
      "specCharacteristic": [
        {
          "@type": "CharacteristicSpecification",
          "name": "originatingCountryCode",
          "valueType": "string",
          "description": "ISO 3166 country code of the caller",
          "regex": "^[A-Z]{2}$",
          "minCardinality": 1,
          "maxCardinality": 1
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "originatingPhoneNumber",
          "valueType": "string",
          "description": "International phone number of the caller, must start with
+",
          "regex": "^\\+[1-9]\\d{1,14}$",
          "minCardinality": 1,
          "maxCardinality": 1
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "destinationCountryCode",
          "valueType": "string",
          "description": "ISO 3166 country code of the called party",
          "regex": "^[A-Z]{2}$",
          "minCardinality": 1,
          "maxCardinality": 1
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "destinationPhoneNumber",
          "valueType": "string",
          "description": "International phone number of the called party, must start
with +",
          "regex": "^\\+[1-9]\\d{1,14}$",
          "minCardinality": 1,
          "maxCardinality": 1
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "duration",
```

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```
          "valueType": "integer",
          "description": "Total duration of the call, in seconds.",
          "minCardinality": 1,
          "maxCardinality": 1
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "startDateTime",
          "valueType": "date-time",
          "description": "startdate and starttime of the call, in date-time format",
          "minCardinality": 1,
          "maxCardinality": 1
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "endDateTime",
          "valueType": "date-time",
          "description": "EndDate and EndTime of the call",
          "minCardinality": 1,
          "maxCardinality": 1
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "callType",
          "valueType": "string",
          "description": "Type of the call : outgoing or incoming",
          "minCardinality": 1,
          "maxCardinality": 1,
          "characteristicValueSpecification": [
            {
              "@type": "StringCharacteristicValueSpecification",
              "valueType": "string",
              "value": "outgoing"
            },
            {
              "@type": "StringCharacteristicValueSpecification",
              "valueType": "string",
              "value": "incoming"
            }
          ]
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "callStatus",
          "valueType": "string",
          "description": "Status of the call : failed or completed",
          "minCardinality": 1,
          "maxCardinality": 1,
          "characteristicValueSpecification": [
            {
              "@type": "StringCharacteristicValueSpecification",
              "valueType": "string",
              "value": "failed"
            },
            {
              "@type": "StringCharacteristicValueSpecification",
              "valueType": "string",
              "value": "completed"
            }
          ]
        }
      ]
    }
```


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```
  }
}
```

State Change Event

Creating a ProductUsageSpecificationStateChangeEvent event

```
Content-Type: application/json
{
  "@type": "ProductUsageSpecificationStateChangeEvent",
  "@baseType": "Event",
  "domain": "Product Usage",
  "timeOccurred": "2022-12-01T10:00:00Z",
  "eventId": "d19dcc48-c515-49c1-bdda-52db6f1952d1",
  "eventType": "ProductUsageSpecificationStateChangeEvent",
  "event": {
    "productUsageSpecification": {
      "@type": "ProductUsageSpecification",
      "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
      "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
      "lifecycleStatus": "obsolete"
    }
  }
}
```

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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on ProductUsageSpecification

List or find ProductUsageSpecification objects

`GET /productUsageSpecification?fields=…`

Description

This operation list ProductUsageSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving a list of product usage specification with version 2.5.0

Request

```
GET /productUsageSpecification?version=2.5.0
Content-Type: application/json
```

Response

```
200
[
```

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```
  {
    "@type": "ProductUsageSpecification",
    "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
    "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
    "name": "InternationalVoice",
    "description": "Specification for international voice calls usage",
    "version": "2.5.0",
    "lastUpdate": "2022-05-20T19:00:00Z",
    "lifecycleStatus": "inDesign",
    "specCharacteristic": [
      {
        "@type": "CharacteristicSpecification",
        "name": "originatingCountryCode",
        "valueType": "string",
        "description": "ISO 3166 country code of the caller",
        "regex": "^[A-Z]{2}$",
        "minCardinality": 1,
        "maxCardinality": 1
      },
      {
        "@type": "CharacteristicSpecification",
        "name": "originatingPhoneNumber",
        "valueType": "string",
        "description": "International phone number of the caller, must start with
+",
        "regex": "^\\+[1-9]\\d{1,14}$",
        "minCardinality": 1,
        "maxCardinality": 1
      },
      {
        "@type": "CharacteristicSpecification",
        "name": "destinationCountryCode",
        "valueType": "string",
        "description": "ISO 3166 country code of the called party",
        "regex": "^[A-Z]{2}$",
        "minCardinality": 1,
        "maxCardinality": 1
      },
      {
        "@type": "CharacteristicSpecification",
        "name": "destinationPhoneNumber",
        "valueType": "string",
        "description": "International phone number of the called party, must start
with +",
        "regex": "^\\+[1-9]\\d{1,14}$",
        "minCardinality": 1,
        "maxCardinality": 1
      },
      {
        "@type": "CharacteristicSpecification",
        "name": "duration",
        "valueType": "integer",
        "description": "Total duration of the call, in seconds.",
        "minCardinality": 1,
        "maxCardinality": 1
      },
      {
        "@type": "CharacteristicSpecification",
        "name": "startDateTime",
        "valueType": "date-time",
        "description": "startdate and starttime of the call, in date-time format",
```


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```
        "minCardinality": 1,
        "maxCardinality": 1
      },
      {
        "@type": "CharacteristicSpecification",
        "name": "endDateTime",
        "valueType": "date-time",
        "description": "enddate and endtime of the call, in date-time format",
        "minCardinality": 1,
        "maxCardinality": 1
      },
      {
        "@type": "CharacteristicSpecification",
        "name": "callType",
        "valueType": "string",
        "description": "Type of the call : outgoing or incoming",
        "minCardinality": 1,
        "maxCardinality": 1,
        "characteristicValueSpecification": [
          {
            "@type": "StringCharacteristicValueSpecification",
            "valueType": "string",
            "value": "outgoing"
          },
          {
            "@type": "StringCharacteristicValueSpecification",
            "valueType": "string",
            "value": "incoming"
          }
        ]
      },
      {
        "@type": "CharacteristicSpecification",
        "name": "callStatus",
        "valueType": "string",
        "description": "Status of the call : failed or completed",
        "minCardinality": 1,
        "maxCardinality": 1,
        "characteristicValueSpecification": [
          {
            "@type": "StringCharacteristicValueSpecification",
            "valueType": "string",
            "value": "failed"
          },
          {
            "@type": "StringCharacteristicValueSpecification",
            "valueType": "string",
            "value": "completed"
          }
        ]
      }
    ],
    "validFor": {
      "startDateTime": "2022-05-20T19:00:00Z",
      "endDateTime": "2099-01-11T00:00:00Z"
    },
    "productSpecification": [
      {
        "@type": "ProductSpecificationRef",
        "@referredType": "ProductSpecification",
        "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productSpecification/cbfa3598-2276-4b35-8f7f-
3252d9b8f397",
```


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```
        "id": "cbfa3598-2276-4b35-8f7f-3252d9b8f397",
        "name": "FixCommunication",
        "version": "1.0"
      }
    ],
    "serviceUsageSpecification": [
      {
        "@type": "ServiceUsageSpecificationRef",
        "@referredType": "ServiceUsageSpecification",
        "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsageSpecification/6d414e5c-6e49-4651-b4a9-
64f698bda943",
        "id": "6d414e5c-6e49-4651-b4a9-64f698bda943",
        "name": "InternationalVoice"
      }
    ]
  }
]
```


Retrieves a ProductUsageSpecification by ID

`GET /productUsageSpecification/{id}?fields=…`

Description

This operation retrieves a ProductUsageSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving a product usage specification with its id c2d6642b-59be-4cd1-800d-f3920dd4bc7f

Request

```
GET /productUsageSpecification/c2d6642b-59be-4cd1-800d-f3920dd4bc7f
Content-Type: application/json
```

Response


```
200
{
  "@type": "ProductUsageSpecification",
  "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
  "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
  "name": "InternationalVoice",
  "description": "Specification for international voice calls usage",
  "version": "2.5.0",
  "lastUpdate": "2022-05-20T19:00:00Z",
  "lifecycleStatus": "inDesign",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
```


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```
      "name": "originatingCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the caller",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the caller, must start with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the called party",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the called party, must start
with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "duration",
      "valueType": "integer",
      "description": "Total duration of the call, in seconds.",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "startDateTime",
      "valueType": "date-time",
      "description": "startdate and starttime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "endDateTime",
      "valueType": "date-time",
      "description": "enddate and endtime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callType",
      "valueType": "string",
```


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```
      "description": "Type of the call : outgoing or incoming",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "outgoing"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "incoming"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callStatus",
      "valueType": "string",
      "description": "Status of the call : failed or completed",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "failed"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "completed"
        }
      ]
    }
  ],
  "validFor": {
    "startDateTime": "2022-05-20T19:00:00Z",
    "endDateTime": "2099-01-11T00:00:00Z"
  },
  "productSpecification": [
    {
      "@type": "ProductSpecificationRef",
      "@referredType": "ProductSpecification",
      "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productSpecification/cbfa3598-2276-4b35-8f7f-
3252d9b8f397",
      "id": "cbfa3598-2276-4b35-8f7f-3252d9b8f397",
      "name": "FixCommunication"
    }
  ],
  "serviceUsageSpecification": [
    {
      "@type": "ServiceUsageSpecificationRef",
      "@referredType": "ServiceUsageSpecification",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsageSpecification/6d414e5c-6e49-4651-b4a9-
64f698bda943",
      "id": "6d414e5c-6e49-4651-b4a9-64f698bda943",
      "name": "InternationalVoice"
    }
  ]
```


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```
}
```

Creates a ProductUsageSpecification

POST /productUsageSpecification?fields=…

Description

This operation creates a ProductUsageSpecification entity.

Mandatory Attributes

Mandatory Attributes	Rule
lifecycleStatus	
name	
@type	M

Usage samples

Creating a product usage specification resource

Request

```
POST /productUsageSpecification
Content-Type: application/json
{
  "@type": "ProductUsageSpecification",
  "name": "InternationalVoice",
  "description": "Specification for international voice calls usage",
  "version": "2.5.0",
  "lifecycleStatus": "inDesign",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the caller",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the caller, must start with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the called party",
```

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```
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the called party, must start
with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "duration",
      "valueType": "integer",
      "description": "Total duration of the call, in seconds.",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "startDateTime",
      "valueType": "date-time",
      "description": "startdate and starttime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "endDateTime",
      "valueType": "date-time",
      "description": "enddate and endtime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callType",
      "valueType": "string",
      "description": "Type of the call : outgoing or incoming",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "outgoing"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "incoming"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callStatus",
      "valueType": "string",
      "description": "Status of the call : failed or completed",
```


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```
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "failed"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "completed"
        }
      ]
    }
  ],
  "validFor": {
    "startDateTime": "2022-02-20T17:00:00Z",
    "endDateTime": "2099-01-11T00:00:00Z"
  },
  "productSpecification": [
    {
      "@type": "ProductSpecificationRef",
      "@referredType": "ProductSpecification",
      "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productSpecification/cbfa3598-2276-4b35-8f7f-
3252d9b8f397",
      "id": "cbfa3598-2276-4b35-8f7f-3252d9b8f397",
      "name": "FixCommunication",
      "version": "1.0"
    }
  ],
  "serviceUsageSpecification": [
    {
      "@type": "ServiceUsageSpecificationRef",
      "@referredType": "ServiceUsageSpecification",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsageSpecification/6d414e5c-6e49-4651-b4a9-
64f698bda943",
      "id": "6d414e5c-6e49-4651-b4a9-64f698bda943",
      "name": "InternationalVoice"
    }
  ]
}
```


Response


```
201
{
  "@type": "ProductUsageSpecification",
  "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
  "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
  "name": "InternationalVoice",
  "description": "Specification for international voice calls usage",
  "version": "2.5.0",
  "lastUpdate": "2022-05-20T19:00:00Z",
  "lifecycleStatus": "inDesign",
  "specCharacteristic": [
```


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```
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the caller",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the caller, must start with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the called party",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the called party, must start
with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "duration",
      "valueType": "integer",
      "description": "Total duration of the call, in seconds.",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "startDateTime",
      "valueType": "date-time",
      "description": "startdate and starttime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "endDateTime",
      "valueType": "date-time",
      "description": "enddate and endtime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
```


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```
      "name": "callType",
      "valueType": "string",
      "description": "Type of the call : outgoing or incoming",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "outgoing"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "incoming"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callStatus",
      "valueType": "string",
      "description": "Status of the call : failed or completed",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "failed"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "completed"
        }
      ]
    }
  ],
  "validFor": {
    "startDateTime": "2022-05-20T19:00:00Z",
    "endDateTime": "2099-01-11T00:00:00Z"
  },
  "productSpecification": [
    {
      "@type": "ProductSpecificationRef",
      "@referredType": "ProductSpecification",
      "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productSpecification/cbfa3598-2276-4b35-8f7f-
3252d9b8f397",
      "id": "cbfa3598-2276-4b35-8f7f-3252d9b8f397",
      "name": "FixCommunication",
      "version": "1.0"
    }
  ],
  "serviceUsageSpecification": [
    {
      "@type": "ServiceUsageSpecificationRef",
      "@referredType": "ServiceUsageSpecification",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsageSpecification/6d414e5c-6e49-4651-b4a9-
64f698bda943",
      "id": "6d414e5c-6e49-4651-b4a9-64f698bda943",
```


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```
        "name": "InternationalVoice"
      }
    ]
  }
```

Updates partially a ProductUsageSpecification

PATCH /productUsageSpecification/{id}?fields=…

Description

This operation allows partial updates of a ProductUsageSpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
lastUpdate	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
attachment	
description	
lifecycleStatus	
name	
productSpecification	
serviceUsageSpecification	
specCharacteristic	
targetProductUsageSchema	
validFor	
version	

Usage samples

Product usage specification update example with json content type

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Request

```
PATCH /productUsageSpecification/c2d6642b-59be-4cd1-800d-f3920dd4bc7f
Content-Type: application/json
{
  "@type": "ProductUsageSpecification",
  "version": "3.2"
}
```

Response

```
200
{
  "@type": "ProductUsageSpecification",
  "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
  "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
  "name": "InternationalVoice",
  "description": "Specification for international voice calls usage",
  "version": "3.2",
  "lastUpdate": "2022-05-30T10:00:00Z",
  "lifecycleStatus": "inDesign",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the caller",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the caller, must start with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the called party",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the called party, must start
with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
```

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```
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "duration",
      "valueType": "integer",
      "description": "Total duration of the call, in seconds.",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "startDateTime",
      "valueType": "date-time",
      "description": "startdate and starttime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "endDateTime",
      "valueType": "date-time",
      "description": "enddate and endtime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callType",
      "valueType": "string",
      "description": "Type of the call : outgoing or incoming",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "outgoing"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "incoming"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callStatus",
      "valueType": "string",
      "description": "Status of the call : failed or completed",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "failed"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "completed"
```


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```
        }
      ]
    }
  ],
  "validFor": {
    "startDateTime": "2022-05-20T19:00:00Z",
    "endDateTime": "2099-01-11T00:00:00Z"
  },
  "productSpecification": [
    {
      "@type": "ProductSpecificationRef",
      "@referredType": "ProductSpecification",
      "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productSpecification/cbfa3598-2276-4b35-8f7f-
3252d9b8f397",
      "id": "cbfa3598-2276-4b35-8f7f-3252d9b8f397",
      "name": "FixCommunication",
      "version": "1.0"
    }
  ],
  "serviceUsageSpecification": [
    {
      "@type": "ServiceUsageSpecificationRef",
      "@referredType": "ServiceUsageSpecification",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsageSpecification/6d414e5c-6e49-4651-b4a9-
64f698bda943",
      "id": "6d414e5c-6e49-4651-b4a9-64f698bda943",
      "name": "InternationalVoice"
    }
  ]
}
```


Here is an example of a request for patching a ProductUsageSpecification

Request

```
PATCH /productUsageSpecification/c2d6642b-59be-4cd1-800d-f3920dd4bc7f
Content-Type: application/merge-patch+json
{
  "@type": "ProductUsageSpecification",
  "version": "3.2"
}
```

Response


```
200
{
  "@type": "ProductUsageSpecification",
  "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
  "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
  "name": "InternationalVoice",
  "description": "Specification for international voice calls usage",
  "version": "3.2",
  "lastUpdate": "2022-05-30T10:00:00Z",
```


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```
  "lifecycleStatus": "inDesign",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the caller",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the caller, must start with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the called party",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the called party, must start
with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "duration",
      "valueType": "integer",
      "description": "Total duration of the call, in seconds.",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "startDateTime",
      "valueType": "date-time",
      "description": "startdate and starttime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "endDateTime",
      "valueType": "date-time",
      "description": "enddate and endtime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
```


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```
    {
      "@type": "CharacteristicSpecification",
      "name": "callType",
      "valueType": "string",
      "description": "Type of the call : outgoing or incoming",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "outgoing"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "incoming"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callStatus",
      "valueType": "string",
      "description": "Status of the call : failed or completed",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "failed"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "completed"
        }
      ]
    }
  ],
  "validFor": {
    "startDateTime": "2022-05-20T19:00:00Z",
    "endDateTime": "2099-01-11T00:00:00Z"
  },
  "productSpecification": [
    {
      "@type": "ProductSpecificationRef",
      "@referredType": "ProductSpecification",
      "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productSpecification/cbfa3598-2276-4b35-8f7f-
3252d9b8f397",
      "id": "cbfa3598-2276-4b35-8f7f-3252d9b8f397",
      "name": "FixCommunication",
      "version": "1.0"
    }
  ],
  "serviceUsageSpecification": [
    {
      "@type": "ServiceUsageSpecificationRef",
      "@referredType": "ServiceUsageSpecification",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsageSpecification/6d414e5c-6e49-4651-b4a9-
```


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```
64f698bda943",
      "id": "6d414e5c-6e49-4651-b4a9-64f698bda943",
      "name": "InternationalVoice"
    }
  ]
}
```

Product usage specification update example with json-patch+json content type

Request

```
PATCH /productUsageSpecification/c2d6642b-59be-4cd1-800d-f3920dd4bc7f
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "/version",
    "value": "3.0"
  }
]
```

Response

```
200
{
  "@type": "ProductUsageSpecification",
  "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
  "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
  "name": "InternationalVoice",
  "description": "Specification for international voice calls usage",
  "version": "3.0",
  "lastUpdate": "2022-05-30T10:00:00Z",
  "lifecycleStatus": "inDesign",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the caller",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the caller, must start with +",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the called party",
```

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```
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the called party, must start
with +",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "duration",
      "valueType": "integer",
      "description": "Total duration of the call, in seconds.",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "startDateTime",
      "valueType": "date-time",
      "description": "startdate and starttime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "endDateTime",
      "valueType": "date-time",
      "description": "enddate and endtime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callType",
      "valueType": "string",
      "description": "Type of the call : outgoing or incoming",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "outgoing"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "incoming"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callStatus",
      "valueType": "string",
      "description": "Status of the call : failed or completed",
      "minCardinality": 1,
      "maxCardinality": 1,
```


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```
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "failed"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "completed"
        }
      ]
    }
  ],
  "validFor": {
    "startDateTime": "2022-05-20T19:00:00Z",
    "endDateTime": "2099-01-11T00:00:00Z"
  },
  "productSpecification": [
    {
      "@type": "ProductSpecificationRef",
      "@referredType": "ProductSpecification",
      "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productSpecification/cbfa3598-2276-4b35-8f7f-
3252d9b8f397",
      "id": "cbfa3598-2276-4b35-8f7f-3252d9b8f397",
      "name": "FixCommunication",
      "version": "1.0"
    }
  ],
  "serviceUsageSpecification": [
    {
      "@type": "ServiceUsageSpecificationRef",
      "@referredType": "ServiceUsageSpecification",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsageSpecification/6d414e5c-6e49-4651-b4a9-
64f698bda943",
      "id": "6d414e5c-6e49-4651-b4a9-64f698bda943",
      "name": "InternationalVoice"
    }
  ]
}
```


Product usage specification update example with json-patch-query+json content type

Request

```
PATCH /productUsageSpecification/c2d6642b-59be-4cd1-800d-f3920dd4bc7f
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path": "/specCharacteristic/description?name=endDateTime",
    "value": "EndDate and EndTime of the call"
  }
]
```

Response

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```
200
{
  "@type": "ProductUsageSpecification",
  "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
  "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
  "name": "InternationalVoice",
  "description": "Specification for international voice calls usage",
  "version": "2.5.0",
  "lastUpdate": "2022-05-30T10:00:00Z",
  "lifecycleStatus": "inDesign",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the caller",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "originatingPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the caller, must start with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationCountryCode",
      "valueType": "string",
      "description": "ISO 3166 country code of the called party",
      "regex": "^[A-Z]{2}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "destinationPhoneNumber",
      "valueType": "string",
      "description": "International phone number of the called party, must start
with +",
      "regex": "^\\+[1-9]\\d{1,14}$",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "duration",
      "valueType": "integer",
      "description": "Total duration of the call, in seconds.",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "startDateTime",
      "valueType": "date-time",
```

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```
      "description": "startdate and starttime of the call, in date-time format",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "endDateTime",
      "valueType": "date-time",
      "description": "EndDate and EndTime of the call",
      "minCardinality": 1,
      "maxCardinality": 1
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callType",
      "valueType": "string",
      "description": "Type of the call : outgoing or incoming",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "outgoing"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "incoming"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "callStatus",
      "valueType": "string",
      "description": "Status of the call : failed or completed",
      "minCardinality": 1,
      "maxCardinality": 1,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "failed"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "completed"
        }
      ]
    }
  ],
  "validFor": {
    "startDateTime": "2022-05-20T19:00:00Z",
    "endDateTime": "2099-01-11T00:00:00Z"
  },
  "productSpecification": [
    {
      "@type": "ProductSpecificationRef",
      "@referredType": "ProductSpecification",
      "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productSpecification/cbfa3598-2276-4b35-8f7f-
```


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```
3252d9b8f397",
      "id": "cbfa3598-2276-4b35-8f7f-3252d9b8f397",
      "name": "FixCommunication",
      "version": "1.0"
    }
  ],
  "serviceUsageSpecification": [
    {
      "@type": "ServiceUsageSpecificationRef",
      "@referredType": "ServiceUsageSpecification",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsageSpecification/6d414e5c-6e49-4651-b4a9-
64f698bda943",
      "id": "6d414e5c-6e49-4651-b4a9-64f698bda943",
      "name": "InternationalVoice"
    }
  ]
}
```


Deletes a ProductUsageSpecification

`DELETE /productUsageSpecification/{id}`

Description

This operation deletes a ProductUsageSpecification entity.

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release Notes

v5.0.0

- TMF767 first stable version
- productUsageSpecification is an evolution of usageSpecification from TMF635 Usage Management API v4.0.0 which name change from usageSpecification to productUsageSpecification

schemas changes

- productUsageSpecification : entitySpecRelationship removed (the link with the other specifications are specific attributes)
- productUsageSpecification : productSpecification added to give the reference to the ProductSpecification that to which the ProductUsageSpecification(s) is attached
- productUsageSpecification : serviceUsageSpecification added to give reference to serviceUsageSpecification(s) involved for the creation of this product usage specification.
- productUsageSpecification : relatedParty removed as no general need for it
- productUsageSpecification : constraint removed as no API exposes Constraint entity
- productUsageSpecification : targetEntitySchema replaced by targetProductUsageSchema
- productUsageSpecification : name and lifecycleStatus are required attribute for the creation operation

Release History

Release Number	Date	Release led by:	Description
v5.0 preview	June 2024	Sergei Lukin/Deutsche Telekom	Preview version
v5.0.0	November 2025	Olivier Arnaud / Orange	final upgrades for v5

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